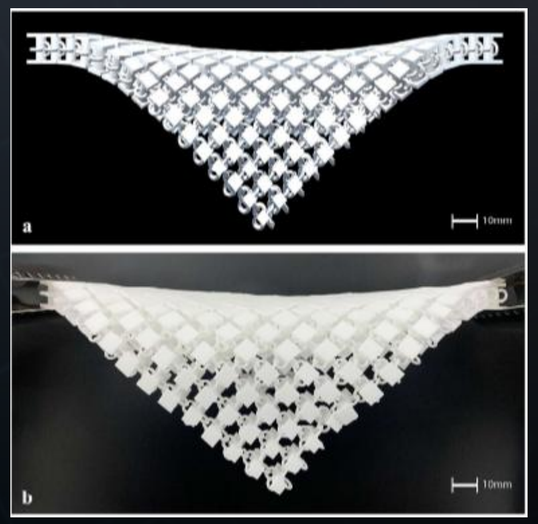
COM6104
Topics in Data Science and Artificial Intelligence

Two Team Members

Idea

Comfy UI

Programmable Materials
i.e. carbon fiber, textile and custom-printed wood grain
(BCC Research, 2020)



Source: Ransley *et al.* (2024)

Idea

Fashion Design Assistant App

Generate the patterns

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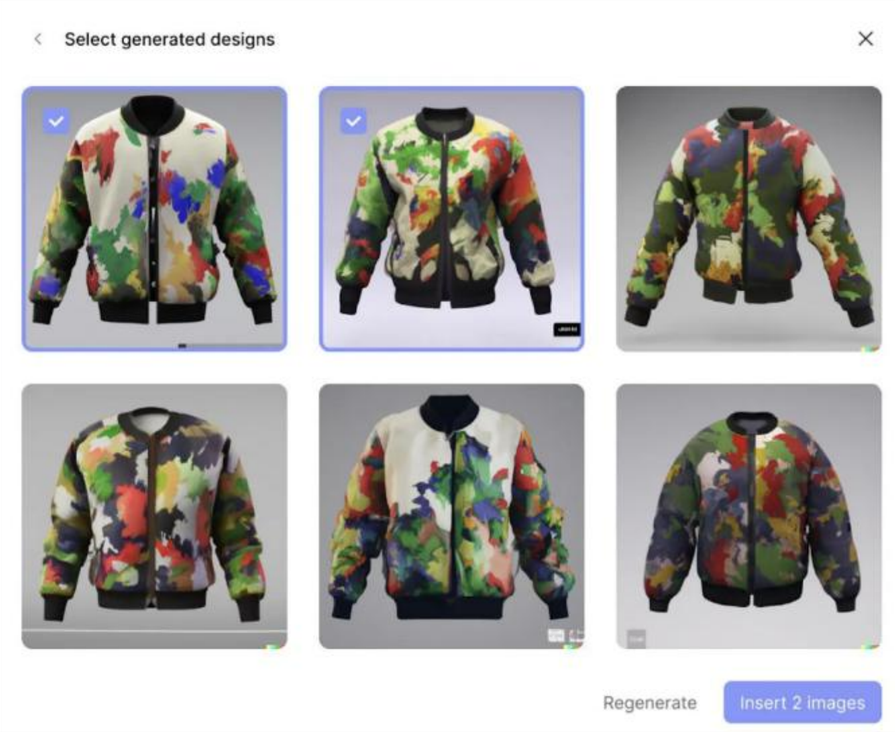
Generate the patterns



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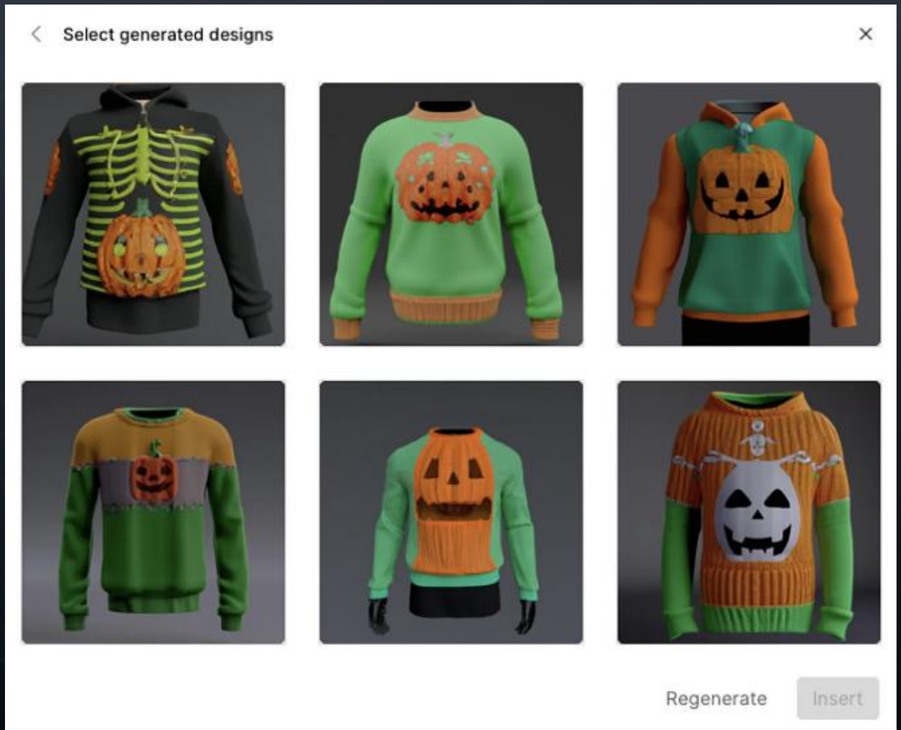


Idea

Fashion Design Assistant App

Build up virtual wardrobe

Idea

Fashion Design Assistant App

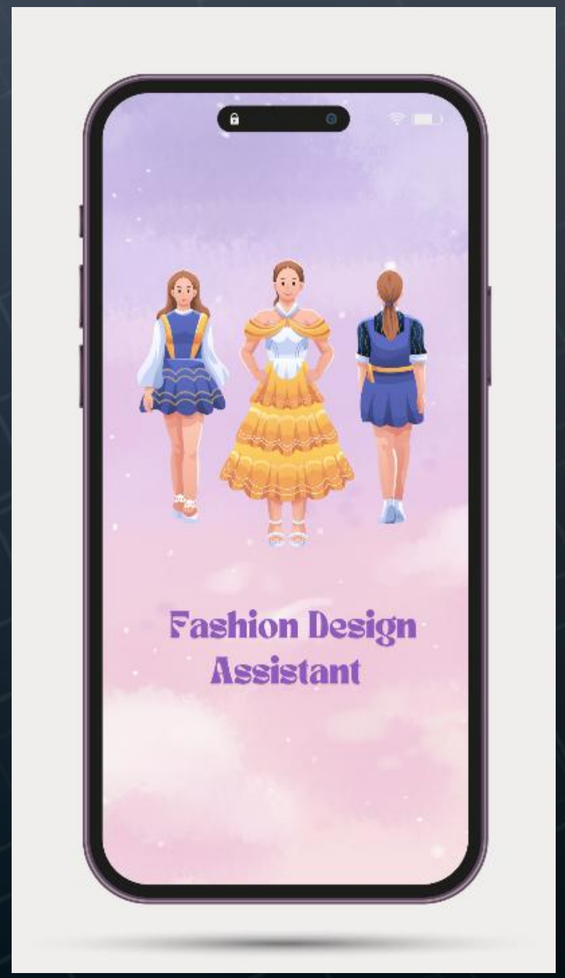
MIX and MATCH

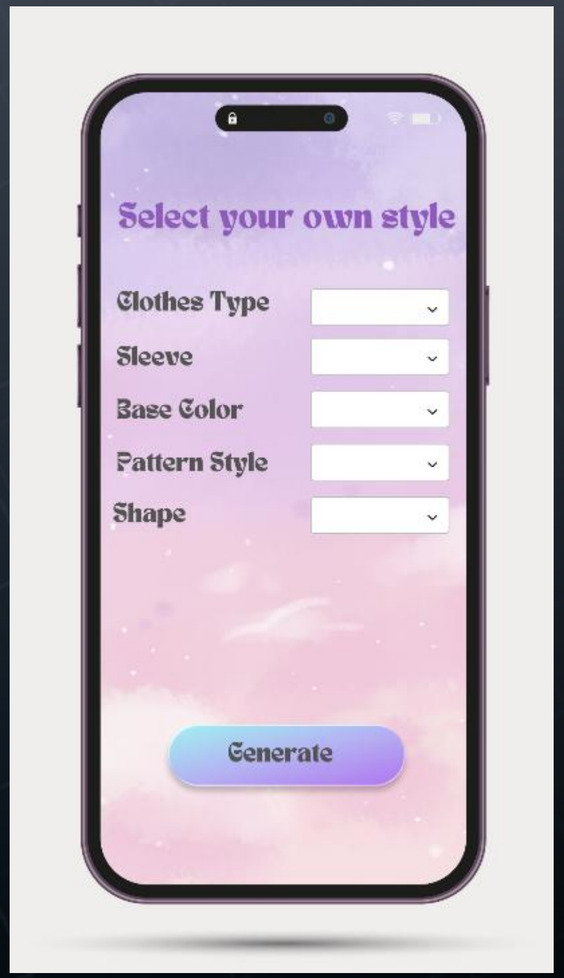
Idea

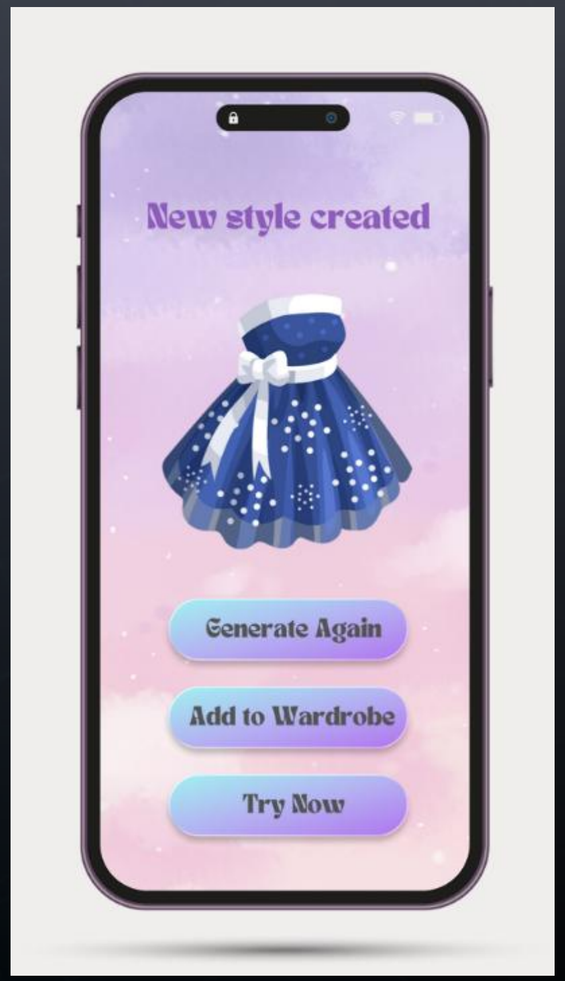
Fashion Design Assistant App

change clothes
by clicking one button

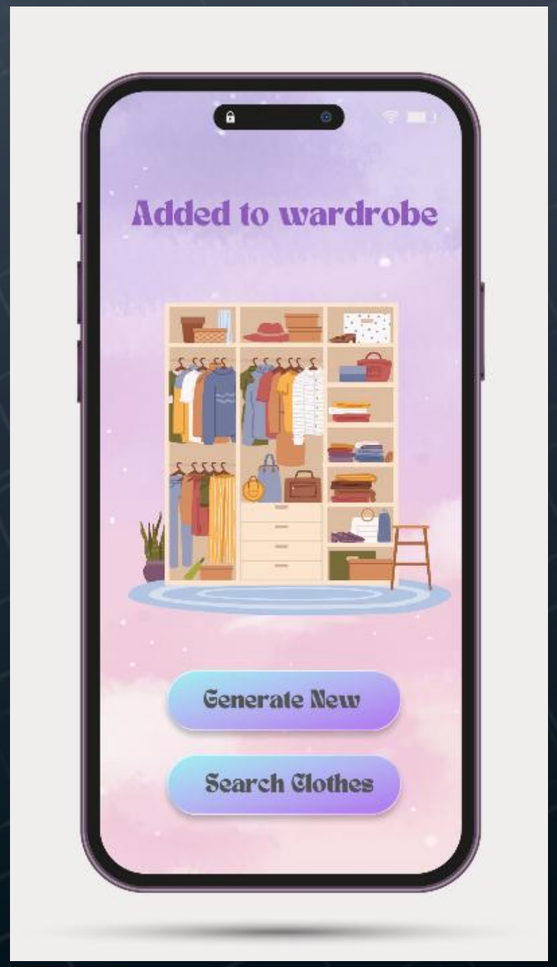
Mock up

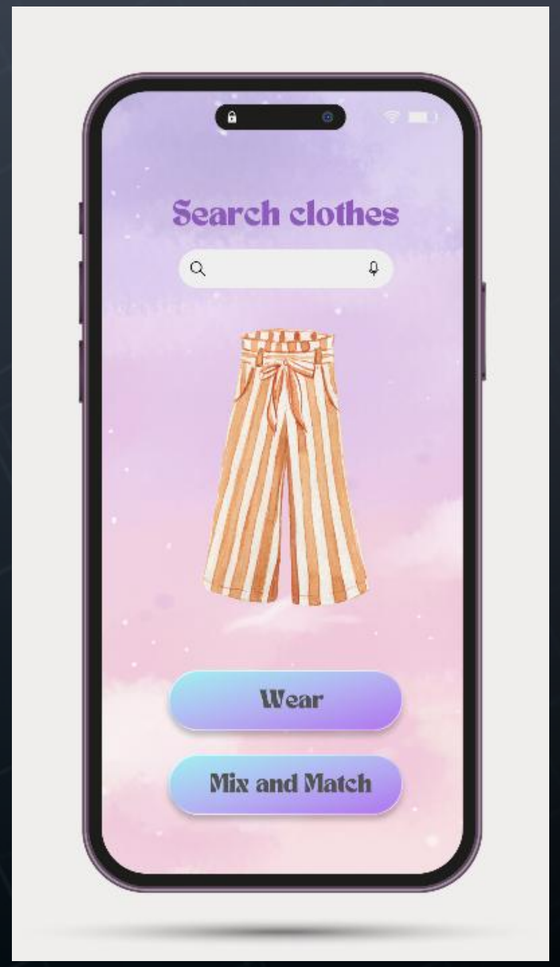

Fashion Design
Assistant

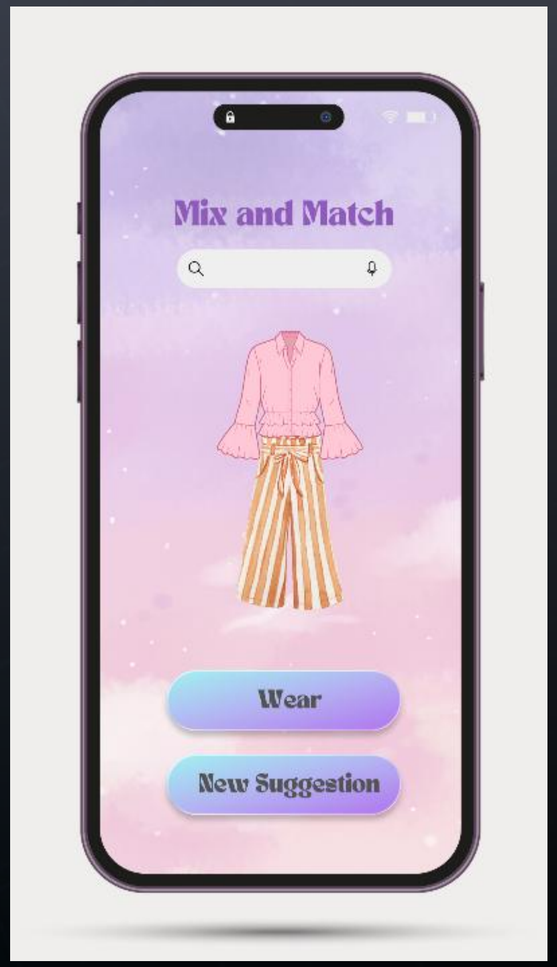

Select your own style
Clothes Type
Sleeve
Base Color
Pattern Style
Shape
Generate


New style created
Generate Again
Add to Wardrobe
Try Now

Mock up


Added to wardrobe
Generate New
Search Clothes


Search clothes
Wear
Mix and Match


Mix and Match
Wear
New Suggestion

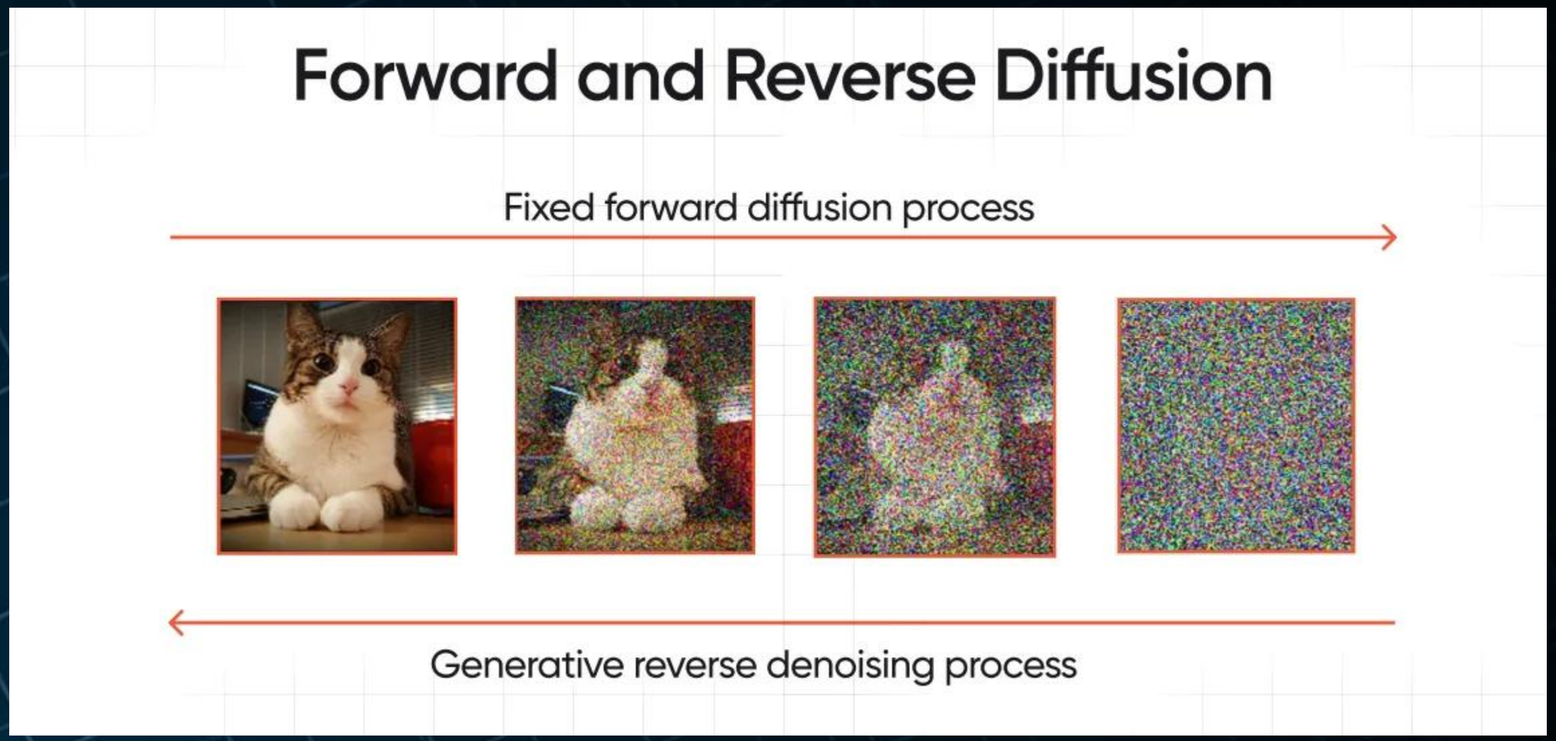
Running Application

Python Container Application
- Handle RESTful request
- Pass to ComfyUI API

- Text to Image request
- Sketch to Image request

1. Install customized model to container. Install ComfyUI API to container.
2. Deploy container application to Cloud platform.
3. Receive GET request via http, generate new image, and send POST results
4. Use Cloud GPU resources for generative AI.
5. Generate new image **on-the-fly** and deliver to mobile app, based on Text to Image prompt, or Sketch to Image prompt.

Diffusion Process



Source: Rishi (2025)

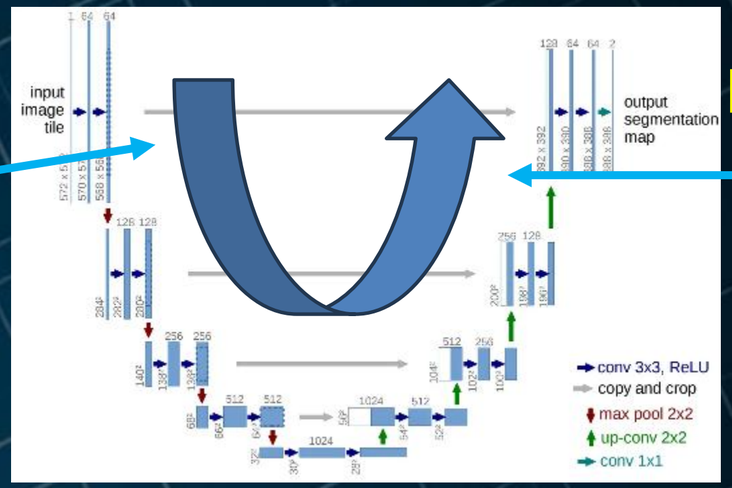
Magic of Stable Diffusion

Diffusion

U-Net (Latent Space)

Encoder

Add Noise



Decoder

Remove Noise

Unrecognized

01 Latent Space

Stable Diffusion encodes the image into Latent Space to make computations more efficient than working with full high-dimensional image data.

02 U-Net

U-Net compresses the latent representation by adding noise, until the equilibrium the image cannot be recognized. U-Net then decode unrecognized image by removing noise.

03 Noise Removal to generate new

During denoising process, U-Net predicts the noise to be removed, gradually fine-tune the latent representation until it becomes a clean image. Noise is based on Gaussian Distribution. In this way, AI generates the new image.

04 Conclusion

Latent space provides the compressed representation of the image, while U-Net processes this representation to perform tasks like denoising, reconstruction, and feature extraction.

Technical Implementation Pipeline

01 Explore Datasets (Fashion Focused)

Visit Hugging Face Hub to search datasets for Fashion like Fashion200k, DeepFashionCatalog or Fashionpedia.

02 Prepare Data

Load directly from Hugging Face “datasets” library in Python.
Filter the dataset with category we want, e.g. “clothes”.

03 Data Pre-processing

Load directly from Hugging Face “datasets” library in Python.
Filter the dataset with category we want, e.g. “clothes”.
Normalize images for model input.

04 Load Pre-Trained Model

Load Stable Diffusion model using “diffusers” library.

05 Train Model (Custom Diffusion)

Incorporate DreamBooth for customized model training

06 Validate Model

Generate sample image outputs to ensure the model understands “shirts” prompts.

07 ComfyUI API

Integrate my customized trained model, and use LoRA model to further fine-tune. ComfyUI can speed up training compared to full fine-tuning.

Run the WebUI with API support. Make RESTful requests to ask the ComfyUI API to generate images and display them on the mobile phone.

ComfyUI Key Concepts

01

Nodes

Each node performs a specific function, such as processing an image, applying a filter, or executing a mathematical operation.

02

Work Flow

Flow represents the connections between nodes, directing the sequence in which different operations are performed.

03

CLIP

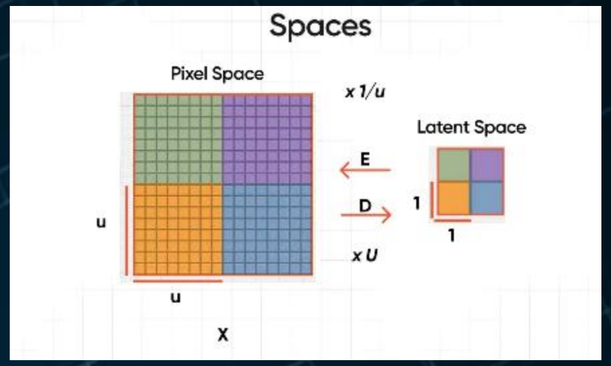
A model to understand text prompts and transform into generative images.

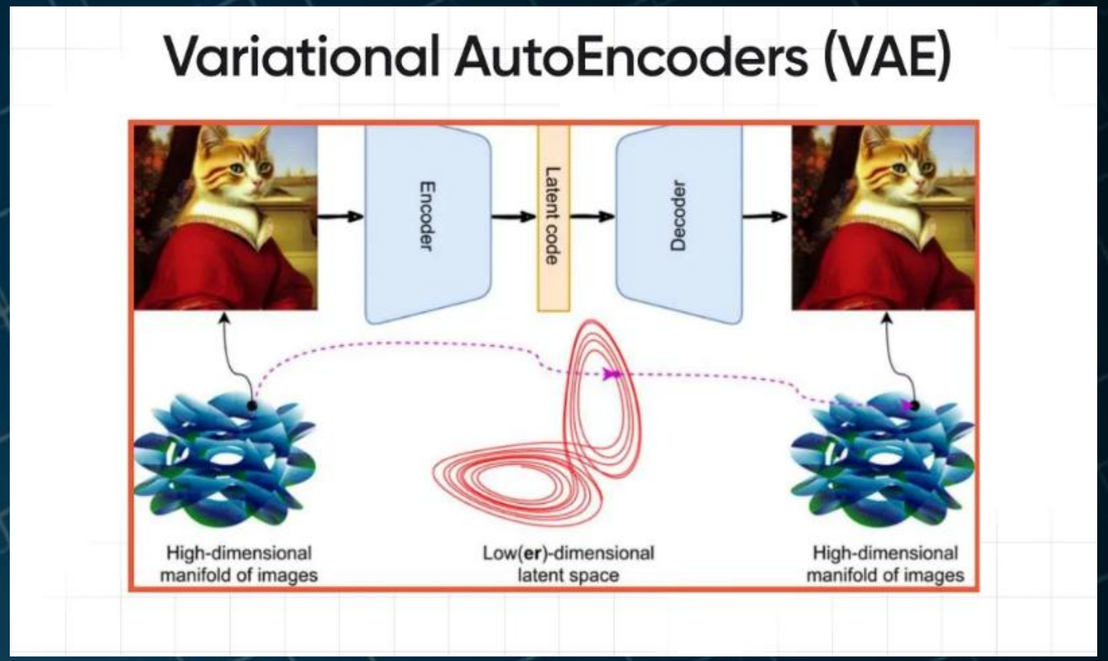
04

Conditioning (KSampler)

Guide the AI to generate images based on specific prompts or constraints, aligning with user expectations.

ComfyUI Key Concepts (cont.)





Source: Rishi (2025)

05 Latent Space

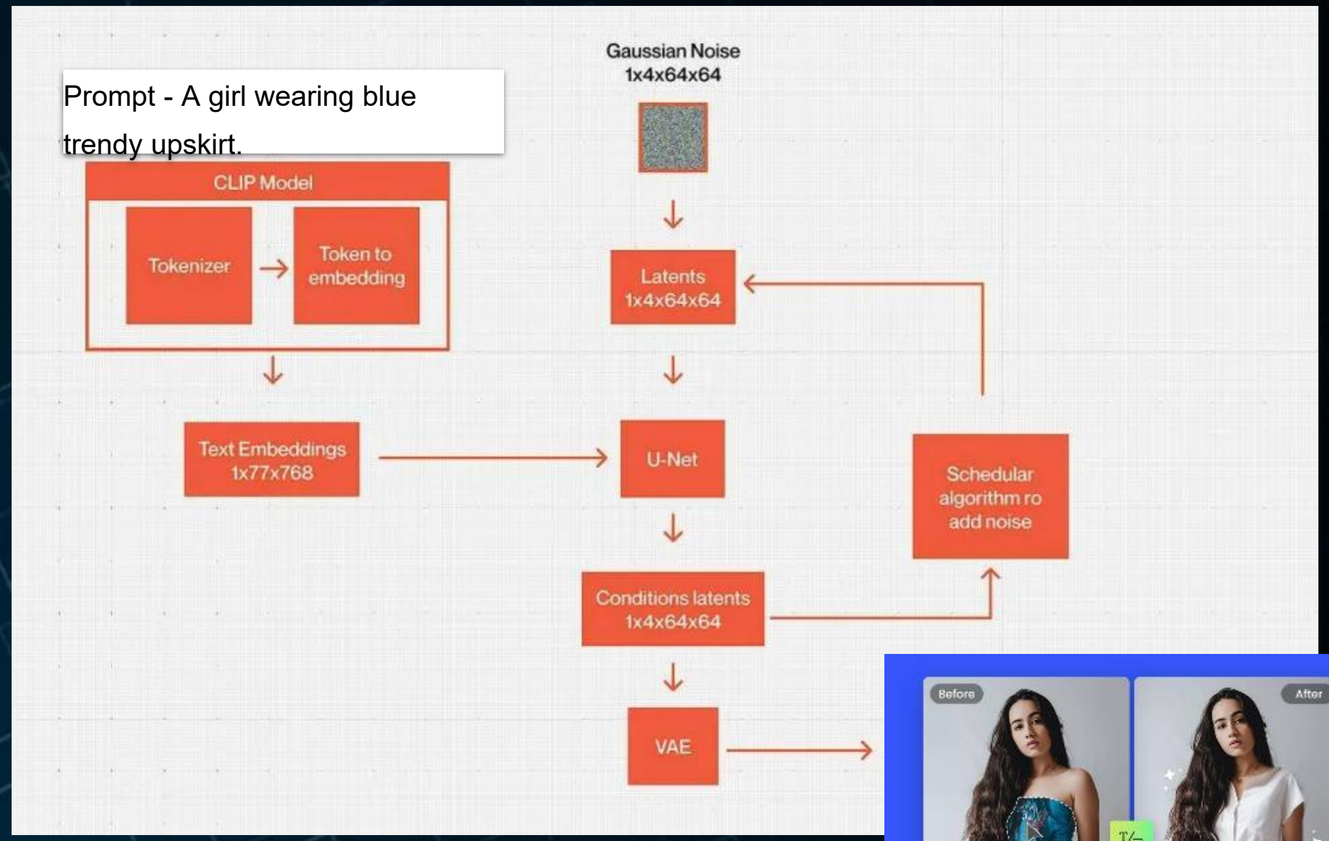
Load directly from Hugging Face “datasets” library in Python.

Filter the dataset with category we want, e.g. “clothes”.

06 VAE (Variational Autoencoder)

Represents the compressed version of an image in a lower-dimensional space, so that AI can generate new images much faster.

How ComfyUI generate



Source: Rishi (2025)

Dream Booth

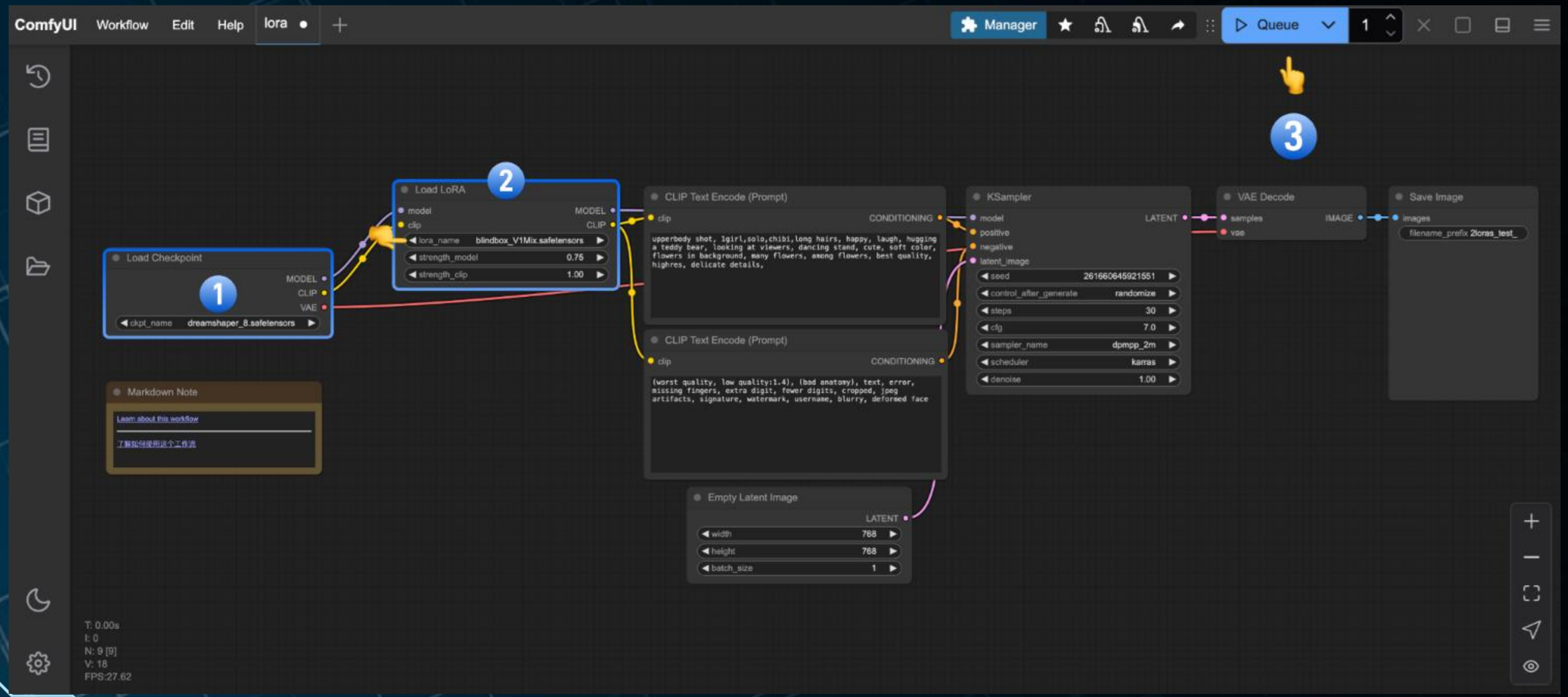
Dreambooth can streamline the process of training our custom diffusion model based on fashion datasets.

However, using Dreambooth could be easy overfitting or underfitting. It is important to tune the learning rate and training steps.

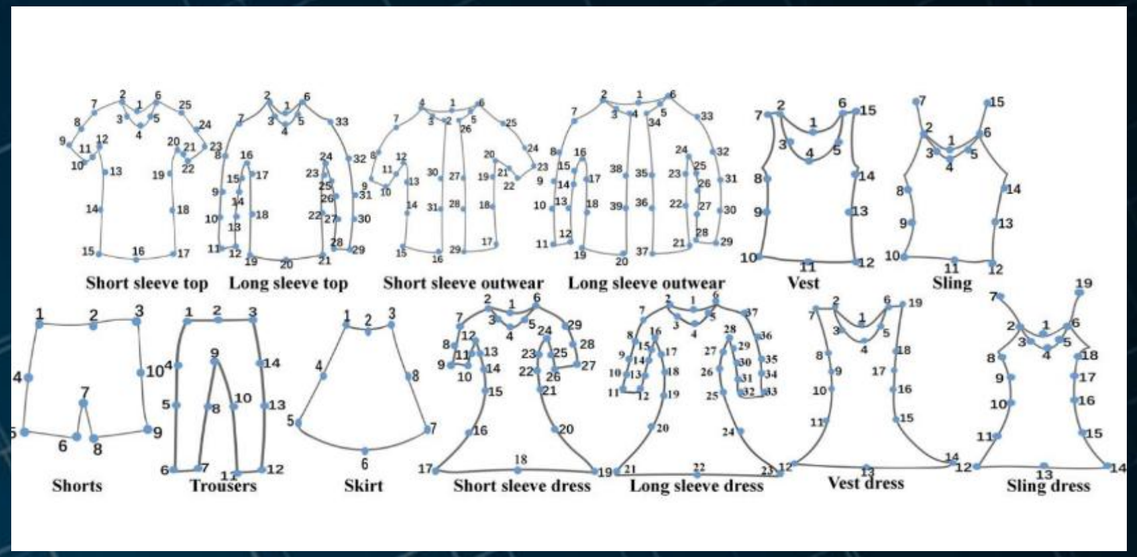
- High Learning rates + too many training steps → The model always mostly generate the same images from training data, no matter what prompt input is.
- Low Learning rates + too few steps → Underfitting; the model cannot generate the image we want.

Lora Model

LoRA, Low-Rank Adaptation introduces low-rank matrices to the our pre-trained model, adjusting only a portion of parameters rather than re-training the whole model, thus achieving efficiency at a lower computational cost.



Sketch to Image



Source: Ge *et al.* (2019)

Git Repository

https://github.com/kyipjp/COM6104-fashion-virtual-try-on

Reference

BCC Research (2020). Programmable Materials. Retrieved from: https://www.bccresearch.com/whitepapers/future-of-programmable-materials.html .

Ge, Y., Zhang, R., Wang, X., Tang, X., & Luo, P. (2019). Deepfashion2: A versatile benchmark for detection, pose estimation, segmentation and re-identification of clothing images. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 5337-5345).

Ransley, M., Partik, C., Porte, E., Ploszajski, A., Jackson, R., Oldfrey, B., Purkiss., D., Purkiss & Miodownik, M. (2024). An introduction to linkage fabrics and their application as programmable materials. Programmable Materials, 2, e5.

Rishi (2025), How Diffusion Model Works. Retrieved from: https://medium.com/@r9sh9r1j/how-diffusion-model-works-08f9863d59b6 .

THANK YOU